

BEAUFORT WEST, South AfricaWMO: **687270**Lat: **32.347S**Lon: **22.573E**Elev: **899**StdP: **90.98**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.4	3.7	-12.8	1.4	19.8	-10.9	1.6	18.7	15.8	18.3	14.1	18.6	2.5	50	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	36.2	16.7	34.6	16.4	33.2	16.1	19.7	27.6	19.0	27.3	18.4	26.9	5.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	14.1	21.8	16.6	13.2	21.7	15.8	12.5	21.4	60.3	27.6	58.0	27.3	55.8	26.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.5	10.6	9.1		-0.5	39.3	0.8	1.1	-1.0	40.1	-1.5	40.7	-1.9	41.3	-2.5	42.0	
				-2.0	21.1	1.2	1.8	-2.9	22.3	-3.6	23.4	-4.3	24.3	-5.2	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	24.2	24.3	22.0	18.5	15.5	12.9	12.6	13.5	15.6	18.3	20.5	22.9		
	DBStd	5.65	3.25	2.94	3.55	4.14	3.93	3.79	4.31	4.25	4.12	4.17	3.80	3.15		
	HDD10.0	53	0	0	0	1	3	13	21	12	3	1	0	0		
	HDD18.3	862	2	1	9	48	106	169	186	162	102	53	21	3		
	CDD10.0	3103	439	399	372	256	174	99	101	119	172	258	314	399		
	CDD18.3	871	182	167	124	53	19	5	7	12	22	52	86	144		
	CDH23.3	9214	2026	1739	1181	469	126	13	22	97	274	647	1028	1593		
	CDH26.7	4001	1011	824	484	141	18	0	1	17	85	251	437	731		
(14) Wind	WSAvg	4.0	4.1	3.9	3.7	3.5	3.8	4.0	4.2	4.2	4.0	4.2	4.2	4.3		
(15) Precipitation	PrecAvg	238	25	33	43	23	14	10	9	15	11	20	24	22		
	PrecMax	424	111	139	138	100	97	26	50	76	51	73	70	106		
	PrecMin	106	0	0	1	1	0	1	0	0	0	0	2	0		
	PrecStd	98	30	35	40	25	21	7	14	20	12	21	18	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	37.6	35.7	32.8	28.8	24.9	25.5	28.7	32.2	35.1	36.0	37.4		
		MCWB	17.4	17.2	16.6	15.7	13.1	11.2	11.0	12.0	13.5	15.0	15.7	16.4		
	2%	DB	36.1	35.4	33.4	30.4	26.5	23.1	23.6	25.9	29.3	32.1	33.5	34.7		
		MCWB	17.1	17.1	16.4	15.1	12.3	10.3	10.3	10.8	12.2	14.3	15.0	16.3		
	5%	DB	34.1	33.5	31.5	28.3	24.6	21.6	21.9	23.8	26.5	29.8	31.3	33.0		
		MCWB	16.7	16.8	16.1	14.5	11.8	9.9	9.8	10.0	11.4	13.7	14.4	16.0		
	10%	DB	32.2	31.6	29.6	26.2	22.9	19.7	20.1	21.5	24.1	27.2	29.1	30.9		
		MCWB	16.3	16.6	15.6	13.8	11.4	9.3	9.3	9.2	10.6	12.9	14.2	15.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	21.0	20.1	18.0	15.2	12.5	12.4	13.8	15.3	17.3	18.2	19.5		
		MCDB	28.5	29.1	27.8	25.6	22.2	21.0	21.6	21.3	26.6	26.0	27.2	28.1		
	2%	WB	19.4	19.8	19.0	16.9	13.9	11.5	11.2	12.3	13.7	16.2	16.9	18.6		
		MCDB	28.4	27.2	26.5	25.1	21.8	19.9	21.0	21.6	23.5	26.1	27.5	27.9		
	5%	WB	18.7	19.2	18.2	16.0	13.1	10.8	10.4	11.2	12.6	15.2	16.2	17.7		
		MCDB	27.9	26.9	26.1	24.8	21.5	19.1	19.9	20.5	23.0	25.2	26.3	27.1		
	10%	WB	18.0	18.5	17.4	15.3	12.4	10.0	9.7	10.2	11.6	14.2	15.3	17.0		
		MCDB	27.3	26.6	25.2	23.4	20.7	18.3	18.7	19.8	22.2	24.5	25.6	26.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	13.5	12.3	10.9	10.1	10.0	10.2	11.1	12.4	13.2	14.0	14.3		
		MCDBR	17.2	16.4	15.1	13.3	12.3	11.4	11.4	13.5	15.9	17.0	17.4	17.3		
		MCWBR	4.0	4.0	4.0	4.2	4.8	4.9	4.9	5.5	6.1	5.1	4.5	4.2		
	5% WB	MCDBR	14.3	12.5	12.1	11.4	10.7	10.2	10.5	11.9	14.0	14.2	14.8	14.0		
		MCWBR	4.0	3.8	3.8	4.1	4.6	4.9	4.9	5.3	5.7	4.9	4.2	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.326	0.315	0.290	0.273	0.262	0.263	0.289	0.306	0.323	0.308	0.316		
	taud		2.562	2.546	2.569	2.641	2.672	2.671	2.653	2.538	2.466	2.458	2.559	2.564		
	Ebn at Noon		1018	993	971	946	917	905	919	931	963	985	1024	1026		
	Edh at Noon		107	105	96	80	69	65	69	87	105	114	107	108		
(44) All-Sky Solar Radiation	RadAvg		8.17	7.33	6.09	4.62	3.55	3.08	3.38	4.29	5.69	6.94	7.99	8.54		
	RadStd		0.29	0.30	0.29	0.20	0.15	0.15	0.17	0.20	0.30	0.27	0.33	0.44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page